Vaghesan Sundaram

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Education

University of Maryland, College Park College Park, MD
B.S. Computer Science (Cybersecurity), B.S. Mathematics, ACES Cybersecurity Minor *Expected December 2027*
Relevant Coursework: Data Structures, Algorithms, Computer Systems, Discrete Structures, Linear Algebra, Statistics, Cryptography, Network Security, Penetration Testing, Technical Writing, Programming Languages

Middlesex College Edison, NJ
A.S. Mathematics, completed during high school *June 2024*

Experience

Cybersecurity Intern June 2026-Present
Redmind Technologies *Remote*

- Identified and documented **7 reproducible findings** across authentication, data exposure, configuration, and dependency risk by combining Semgrep SAST with manual secure code review of a public-facing client application.
- Delivered a developer-facing report and slide briefing that validated evidence, prioritized all 7 findings by severity, and mapped each issue to actionable remediation guidance for developer implementation and client review.
- Developing a PyTorch pipeline with **MuJoCo, robosuite, robomimic, and LeRobot** to measure **training-data and reward-label poisoning** across behavior cloning, transformer, VLA, and offline RL policies.

Information Technology Intern June 2025-August 2025
New Jersey Courts *Toms River, NJ*

- Installed **courtroom recording hardware** to capture legal proceedings and support court recordkeeping.
- Improved IT asset inventory accuracy across **5 departments** (Finance, IT, Civil, Probation, and Operations) by cataloging devices and reconciling physical equipment with tracking records for asset control.

Projects

Agent Least Privilege | *Python, MCP, Pydantic, asyncio, pytest, mypy, Ruff, GitHub Actions* GitHub

- Built a Python **MCP proxy** enforcing launch-time policies for filesystem paths, exact arguments, and call quotas.
- Serialized call quotas, recorded redacted audit events, failed closed on errors, and tracked unknown outcomes.
- Blocked filesystem effects from **12/12 unauthorized proposals** with **10/12 attack tasks** and **24/24 controls** completed in a prior synthetic run.

CTI Claim Provenance | *Python, CISA KEV/CSAF, NVD, CVE, MITRE ATT&CK, Pydantic* GitHub

- Built a **64-question, 24-group point-in-time CTI benchmark**; across 192 evaluations, constrained output raised evidence binding from 83.3% to 86.5% while exact matches fell from 32/96 to 25/96; a 144-response follow-up found no material schema benefit.
- Tracked CISA KEV/CSAF, NVD, CVE, MITRE ATT&CK, and vendor source provenance with SHA-256 hashes.
- Enforced **point-in-time cutoff and source-authority rules** with evidence spans, abstention, and human review.
- Automated typed-schema validation, split isolation, abstention, secret scanning, and privacy-safe aggregate checks.

Keystroke Authentication System | *Python, Flask, SQLite, JWT, bcrypt, CSRF, CORS* GitHub

- Secured 2 Flask services using **3-sample Pearson/MSE keystroke profiles**, 20-character bcrypt fallback, 5-attempt/15-minute rate limiting, 60-second single-use codes, 1-hour JWTs, and atomic profile re-enrollment.
- Hardened login via CSRF, CORS allowlisting, JWT binding, **callback validation**, and server-to-server exchange.

Prophet Prediction Market | *Python, WebSockets, sentence embeddings, clustering* GitHub

- Contributed to an autonomous prediction-market platform integrating **xAI Grok event detection**, OpenRouter trading agents, and Solana smart contracts for on-chain market creation and trading at HackPrinceton.
- Built a 4-route Flask pipeline processing up to **40 X posts/min** by weighting 7 signals and clustering 10+ posts.

Technical Skills

Certification: CompTIA Security+ (SY0-701), March 2026-March 2029
Languages: Python, Java, C++, C, Assembly, Rust
Security: OSINT, Maltego, Shodan, Recon-ng, Maigret, Burp Suite, Nmap, Wireshark, Ghidra, vulnerability triage, TCP/IP, DNS, DHCP, PCAP analysis, Kali Linux, Hashcat, John the Ripper, privilege escalation, exploit development
Tools: Git, GitHub Actions, Docker, Linux, PowerShell, Flask, Pydantic, pytest, Hypothesis, mypy, REST APIs, GDB
Data & ML: PyTorch, MuJoCo, robosuite, robomimic, LeRobot, SQL, SQLite, NumPy, SciPy, scikit-learn